Integrated Framework for Surgery Scheduling Management under Uncertainty: From Strategic to Tactical Planning

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Abstract

Abstract text.

Keywords:

1. Related Works

Surgery scheduling problems have traditionally been classified into three planning levels with different sources of uncertainty: strategic, tactical, and operational (Rahimi and Gandomi, 2020; Cardoen et al., 2010; Wang et al., 2021; Melo and Fogliatto, 2025). At the strategic level, long-term decisions define the desired split of the capacity into specific surgical specialities and specify patient volumes of different medical specialties to be admitted over the planning horizon (Burdett et al., 2023) — a process commonly referred to as Case Mix Planning (CMP). Different objectives have been introduced at this level, such as maximizing profit (Fügener, 2015; Ma and Demeulemeester, 2013) or minimizing long-term costs (Hof et al., 2015; McRae and Brunner, 2020). The tactical level defines a cyclic schedule (often weekly) commonly known as Master Surgery Schedule (MSS) (Bovim et al., 2020; Siqueira et al., 2018; Carneiro et al., 2025). The MSS step allocates time blocks in operating theatres to surgical teams and is reviewed in a medium-term horizon (e.g., monthly or quarterly). Finally, the operational level focuses on the daily operation and derives a Surgical Case Schedule (SCS), which generally assigns individual patients to time blocks previously allocated to specific surgical teams at the tactical level (Vieira et al., 2025; Harris and Claudio, 2022).

Despite the large body of literature in surgery scheduling, there are significant gaps to be bridged. For example, inter-level integration remains limited in scope, often focusing on short-term alignment between surgeries and post-surgical bed occupation, and generally relying on deterministic approaches (Melo and Fogliatto, 2025; Wang et al., 2021; Siqueira et al., 2018). Furthermore, regardless of the planning level, strategies with a focus on guaranteeing waiting time requirements for patients still remain to be proposed. This is probably due to the complex propagation of uncertainty over different planning levels, which demands hybrid approaches that combine mathematical programming, stochastic modelling and queuing theory.

The inherent interconnection between the strategic and tactical levels has motivated unified frameworks that link CMP and MSS decisions (Melo and Fogliatto, 2025). These often adopt sequential or joint optimisation approaches where case mix decisions act as the input to surgery scheduling models (Ma and Demeulemeester, 2013; Fügener, 2015). Modelling approaches typically consist of Mixed Integer Programming (MIP) formulations that focus on maximising hospital profit subject to Operating Theatre (OT) and bed capacity constraints. While these frameworks acknowledge uncertainty, it is generally confined to downstream elements such as length of stay or ward occupancy. Although modelling post-surgical congestion is relevant, such partial treatments remain short-sighted when not complemented by uncertainty in surgery durations and, crucially, by an explicit modelling of dynamic waiting lists across surgical specialities, which constitute the starting point of the entire planning process.

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The perspective can be broadened by incorporating additional uncertainty sources and dynamic elements. For instance, Liu et al. (2019) model elective admissions through a Markov Decision Process (MDP), highlighting the role of waiting-list size in guiding strategic decisions. Despite its conceptual value, the framework disregards essential practical constraints, such as surgeon availability, block scheduling and case-mix heterogeneity, thereby limiting practical applicability. Similarly, McRae and Brunner (2020) demonstrate that explicitly modelling uncertainty and selecting appropriate aggregation levels can significantly improve CMP outcomes and enable joint CMP-MSS optimisation. Nevertheless, waiting-list dynamics remain outside of the modelling scope, implying no full integration of strategic and tactical decisions. Furthermore, their reliance on Sample Average Approximation (SAA) leads to representations that may understate distributional variability. These studies underscore the value of integrating CMP and MSS, yet they also reveal persistent shortcomings: uncertainty is represented only selectively across planning levels, waiting-list requirements are rarely modelled explicitly, and key operational constraints required for real-world implementation remain overlooked.

Uncertainty in surgery duration is another important issue that, when disregarded, may result in unpredictable idle or overtime in operating theatres (Shehadeh and Padman, 2022). Whilst overtime is costly and can lead to cancellations that compromise the quality of care, idle time reflects an inadequate use of operating theatre capacity (Shehadeh and Padman, 2022; Harris and Claudio, 2022; Kayvanfar et al., 2025). With the aim of maximising hospital revenue while preventing excessive overtime, Barz and Rajaram (2015) propose a tactical patient admission and scheduling problem that derives a CMP under multiple resource constraints and stochastic clinical progression. The approach utilises a newsvendor heuristic to reserve capacity for emergency arrivals, and an MDP model to select the patient cohort to be admitted whilst considering probabilistic transitions between clinical states. The model, however, excludes the possibility of cancellations or postponements driven by overtime, as it assumes that the hospital always disposes of the resources to treat each admitted patient. Moreover, by treating resource consumption as deterministic, the framework effectively ignores uncertainty in surgery duration — one of the main drivers of operational instability. As a result, the model underestimates the practical interplay between stochastic surgery durations, cancellations and overtime, limiting its applicability to integrated CMP-MSS decision-making.

Furthermore, Zhang et al. (2020) study a tactical surgery planning with the explicit aim of mitigating overtime risk when allocating elective patients to OTs over an upcoming planning horizon. Their model accounts for stochastic surgery durations and adopts a risk-averse objective that jointly captures both the probability and the expected magnitude of overtime, extending earlier approaches that consider only one of these dimensions (Shylo et al., 2013; Rath et al., 2017). Although they derive the durations from real-world data, eventually they approximate it through a set of discrete scenarios, limiting the full distributional representation. A similar treatment appears in Shehadeh et al. (2026), which introduces a Distributionally Robust Optimisation (DRO) approach to address uncertainty in surgery durations. While their approach minimises worst-case expected cost over an ambiguity set defined by known moments and support, it likewise relies on scenario-based representations. Accurately capturing surgery-duration variability requires a large number of scenarios, substantially increasing computational burden and potentially undermining the practical advantages of DRO and related methods.

These limitations regarding the integration of overtime management and its associated cancellation risks into surgical planning frameworks highlight important research gaps. There remains a need for models that explicitly incorporate cancellation dynamics induced by overtime and that provide a more thorough representation of surgery-duration uncertainty beyond scenario-based approximations. A key challenge which we tackle in this paper is to develop a modular approach to capture full distributional characteristics to build a planning model and use its output to feed an integrated CMP-MSS approach — i.e., not solving an entire model at once.

An additional and often underexplored dimension in integrated surgical planning concerns the incorporation of practical constraints inherent to real-world healthcare settings. Certain surgical activities, such as organ transplantation, impose rigid scheduling requirements that are difficult to reconcile with standard optimisation assumptions (da Silva et al., 2025). Furthermore, public hospitals in developing countries (e.g., Brazil) operate under severe resource constraints, high and volatile demand, and heterogeneous case mixes, all of which exacerbate the complexity of surgical planning (Carneiro et al., 2025; Siqueira et al., 2018; da Silva et al., 2025). These contextual factors challenge the scalability and transferability of many optimisation-based approaches, which often presume stable resources, predictable demand and homogeneous patient populations. Indeed, as noted by Melo and Fogliatto (2025), much of the existing literature is calibrated to relatively small hospitals in developed regions, limiting its relevance for large, resource-constrained systems. Ignoring such operational realities risks producing models that are theoretically stable but difficult to implement in practice, underscoring the need for scheduling frameworks that explicitly accommodate institutional, organisational and contextual constraints.

This paper addresses the identified gaps in the surgical planning literature by proposing an integrated framework that jointly links strategic and tactical decision levels while explicitly accounting for key sources of uncertainty. The framework captures dynamic surgical demand through explicit waiting-list modelling, enabling the enforcement of waiting-time requirements and long-term queue stability (Shortle et al., 2018), and incorporates stochastic surgery durations to properly balance cancellation, idle time, and overtime risks for each surgical session. The proposed approach is grounded in a real-world application at the Clinical Hospital of the University of Campinas (HC-Unicamp) in Brazil, a high-complexity public hospital that provides detailed operational data and a challenging planning environment (HC - Unicamp, 2024b).

The main contributions of this work are as follows:

- A stochastic waiting-list management strategy based on an (R, Q) control policy (Axsäter, 2015) that ensures compliance with waiting-time targets while maintaining long-term queue stability across surgical specialties;
- A cancellation risk management mechanism based on a newsvendor formulation (Arrow et al., 1951), which determines optimal time buffers for the last surgery in each session under duration uncertainty and defines probability distributions for the number of planned surgeries that can be accommodated without incurring overtime;
- A MIP model for operating theatre allocation that: (i) incorporates practical constraints, including those associated with transplant surgeries, while remaining computationally tractable; and (ii) explicitly incorporates the waiting-list management strategy and cancellation risk considerations from the previous items;
- An empirical validation of the integrated framework with data from a large, resource-constrained public hospital, demonstrating its operational relevance and implementability.

By jointly addressing relevant sources of uncertainty within a unified framework, this work introduces an innovative and integrated surgical planning approach that advances the literature by bridging multiple relevant gaps. The approach also bridges the gap between theory and practice by bringing forward a practically grounded and empirically validated framework for long-term management of surgery queues within complex healthcare settings.

2. Problem Description

Hospital das Clínicas Unicamp (HC - Unicamp) is a university hospital located in Campinas, Brazil (HC - Unicamp, 2024b). It is a high-complexity public hospital serving approximately 6.5 million inhabitants from Campinas' metropolitan area and other regions of Brazil that handles around 500,000 patient cases per year. Of these, roughly 10,000 correspond to elective surgical procedures, with the remainder involving outpatient consultations and emergency care (HC - Unicamp, 2024a). In addition to providing healthcare services, HC - Unicamp operates as a teaching and research hospital, training healthcare professionals and conducting medical and interdisciplinary research.

Elective surgical patients follow a complex, multi-stage care pathway (see Figure 1), encompassing pre-operative, intra-operative and post-operative phases. This process begins with waiting-list management, initiated at local medical consultations and coordinated through the regional bed regulation centre (HC - Unicamp, 2024c), a government agency responsible for regulating hospital access across Campinas Metropolitan Region. Patient entry into HC - Unicamp typically occurs via scheduled outpatient appointments, although some emergency cases subsequently become elective and are incorporated into the same planning process. Once referred, patients are assigned to waiting lists according to medical subspecialty. Decisions at this stage directly determine the CMP (i.e., strategic layer), influencing waiting times, speciality-level demand balance and the long-term utilisation of operating theatre capacity (Siqueira et al., 2018).

On the day of surgery, patients enter the upstream unit management phase, where clinical reassessment is performed. Based on clinical requirements, Intensive Care Unit (ICU) support may be required pre- or postoperatively, necessitating prior bed reservation; otherwise, postoperative ward bed availability must be ensured. Patients then undergo pre-anaesthetic evaluation, diagnostic tests and surgical safety checks. If contraindications arise, the procedure is suspended and the patient is either discharged or rescheduled, according to clinical severity. If cleared, the patient

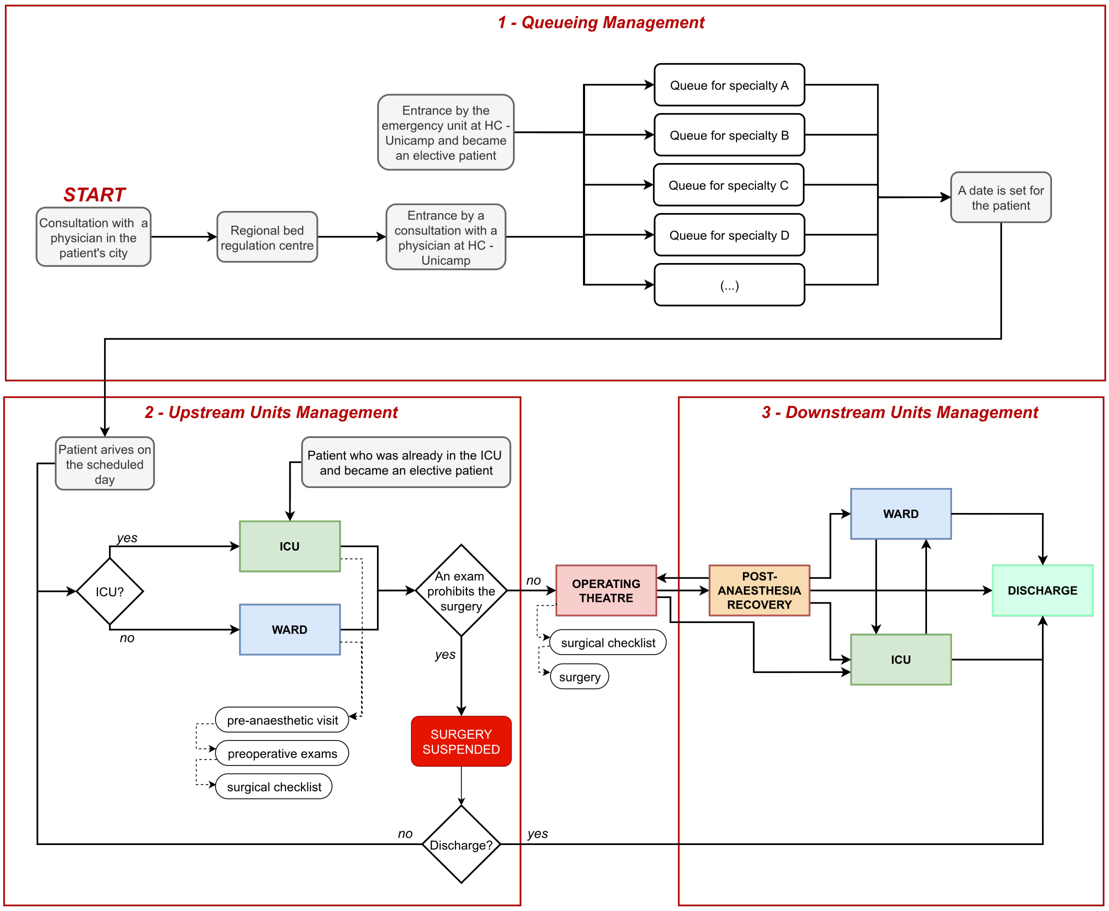

Figure 1: Patient flow at HC - Unicamp.

proceeds to the operating theatre, where the surgical procedure and subsequent theatre cleaning are performed. In the present study, ward and ICU capacity are sufficient and therefore do not constitute binding constraints, allowing the analysis to focus on waiting-list management, uncertainty in surgery duration and operating theatre allocation.

Once in the operating theatre, the patient is managed by a dedicated surgical team and anaesthetist. This stage includes final safety checks, the surgical procedure itself and post-operative cleaning. Here, uncertainty in surgery duration plays a central role, as it directly affects idle time, overtime, and the risk of cancellations. Indeed, it is well known that surgery duration variability has a significant impact on operating theatre utilisation and overall system throughput (e.g., Pandit and Tavare, 2011), therefore meriting explicit consideration in the proposed framework.

Following surgery, patients are transferred to the Post-Anaesthesia Recovery Unit (PARU), with direct ICU transfer occurring only in exceptional cases. After stabilisation, patients are admitted to a ward or ICU bed and subsequently discharged. HC - Unicamp has 15 operating theatres dedicated to elective surgeries: 13 located in the Elective Surgery Centre (ESC) and 2 in the Ambulatory Surgery Centre (ASC), which operates exclusively during morning shifts. These theatres are heterogeneous with respect to infrastructure and equipment, leading to compatibility restrictions across medical subspecialties.

The planning horizon comprises approximately 20 working days per month, each divided into two 6-hour shifts (morning and afternoon). Henceforth, we refer to a theatre shift as a *block*. Hospital policy assigns each block to a single subspecialty, while allowing the same subspecialty to occupy multiple blocks within the same theatre, provided compatibility conditions are met. Sequential allocation is therefore permitted (and preferred by hospital managers), enabling a subspecialty assigned to a morning block to also occupy the corresponding afternoon block.

Each month, operating theatres must be allocated across 40 medical subspecialties. Relevant subsets include: (i) those that mandatorily require two consecutive blocks in the same theatre due to the extended duration of their procedures; (ii) priority subspecialties, which take precedence in allocation decisions in accordance with institutional rules; and (iii) transplant subspecialties, which impose additional operational constraints.

Block allocation feasibility depends mainly on the availability of surgical teams and anaesthetists. A subspecialty can only be assigned to a block if an appropriate team is available on the corresponding day and shift. Moreover, the number of simultaneous allocations requiring anaesthesia cannot exceed the number of anaesthetists available in each period. For example, if 15 anaesthetists are available on a given morning, at most 15 anaesthesia-dependent procedures may be scheduled concurrently.

Day	PERIOD/THEATRE		1	2	3	4	5	6	7	8	9	10	11	12	17	ASC-1	ASC-2
1	MONDAY	MORNING	card sur	tumour orth		procto	uro onco	anaest dor		gast liv	otorhi			vasc			
		AFTERNOON	card sur	tumour orth	pc local	procto	uro onco				otorhi	gast gdp/vb		vasc	vasc local		
2	TUESDAY	MORNING	card sur	foot orth	plast	trauma orth	uro tx		uro tx	gast eed	otorhi	gast eed	neuro	trauma orth			
		AFTERNOON	card sur		plast						otorhi	gast eed	neuro	trauma orth			
3	WEDNESDAY	MORNING	card sur	knee orth	plast	thorax sur	vasc local	paed sur	plast		otorhi	head neck	neuro	trauma orth	vasc		
		AFTERNOON	card sur		plast	thorax sur		paed sur	plast		otorhi	gast obes	neuro		vasc		
4	THURSDAY	MORNING	card sur	eef	adult endo	gast gdp/vb	uro lithiasi	vasc	adult endo	gast liv	otorhi	gast obes	neuro	neuro colu		ophtal	
		AFTERNOON	card sur	pc local	ophtal	gast gdp/vb	uro lithiasi	vasc	otorhi	gast liv	otorhi	gast obes	neuro	neuro colu			
5	FRIDAY	MORNING		paed orth	plast		uro gyneco	paed sur			head neck	head neck	neuro colu			hand orth	
		AFTERNOON		hip orth										trauma orth			

Figure 2: Excerpt from a week's operating theatre schedule.

Figure 2 presents an excerpt from a monthly operating theatre schedule. Rows represent day and shift, and columns correspond to theatres; allocated blocks are colour-coded (blue for procedures requiring anaesthesia and grey otherwise). Blank blocks highlight unallocated slots, reflecting the complexity of the constraint set. Allocations shown in red correspond to transplant surgeries, which require the simultaneous allocation of two theatres within the same day and shift.

Historically, schedules such as those shown in Figure 2 have been manually constructed using only electronic spreadsheets, often requiring more than two full working days. This manual process is time-consuming and does not guarantee optimality with respect to institutional policies. Automating the scheduling process therefore reduces administrative burden and enables the generation of optimal solutions under complex constraints.

In addition to the impact of automating surgery scheduling in a reference hospital in Brazil, this paper provides an approach that is firmly grounded in reality and is therefore adaptable to other general hospitals. By jointly managing waiting-list dynamics, surgery duration uncertainty and operating theatre allocation, our proposed approach provides a coherent and practically grounded solution capable of satisfying waiting-time requirements while mitigating idle time, overtime and cancellation risks.

3. Long-Term Waiting List Management

As discussed in Section 1, CMP constitutes the strategic decision layer responsible for determining the volumes of elective patients to be admitted over a given planning horizon. In contrast to much of the existing literature, which primarily selects these volumes to optimise financial or capacity-related objectives, this work adopts a stability-oriented perspective. We aim to enable decision-makers to impose maximum acceptable waiting times (e.g., six months) and to guarantee, in the long run, that such targets are met without allowing waiting lists to grow unboundedly. This is particularly relevant in the context of public healthcare systems, such as Brazil's Unified Health System (SUS), where there is a legal obligation to provide care to all patients in need. Thus, ensuring equitable access to care and adherence to waiting time standards is paramount.

In line with this objective, this section introduces the theoretical foundations of the modified (R, Q) control policy proposed to manage elective surgery waiting lists under stochastic demand. We first characterise the dynamics of the underlying stochastic process and analyse its long-run behaviour. Subsequently, we derive analytical upper and lower bounds on the batch size Q that ensures compliance with prescribed waiting-time requirements, thereby establishing the strategic input for the integrated planning framework.

3.1. (R, Q) surgery scheduling policy: long-term distributions

Let X_t, $t \geq 0$ be a stochastic process representing an elective surgery queue with stationary and independent increments over discrete planning periods, with X_t representing the number of patients in the system at period $t \geq 0$. Assume that the queue is controlled via a modified (R, Q) policy (e.g., Axsäter, 2015) whereby a batch of $Q \in \mathbb{N}$ surgeries is processed at time period t whenever $X_t \geq R$, $R \geq Q$ at the start of the period, where $\mathbb{N}$ is the set of positive integers. In contrast to the classical (R, Q) policy, which processes as many batches as necessary to bring the inventory below the reorder point (R), our policy can only process a batch at a time and relies on long-term properties to ensure that the process X_t, $t \geq 0$ is stable and guarantees prescribed waiting time requirements.

Let D be a non-negative integer random variable representing the overall demand between two consecutive planning periods, with probability distribution $f_D : \Omega \to [0, 1)$, $\Omega \subset \mathbb{Z}_+$, and cumulative distribution $F_D : \mathbb{Z}_+ \to [0, 1]$ given by

$$F_D(d) = \sum_{l=0}^{d} f_D(l),$$

where $\mathbb{Z}_+$ denotes the set of non-negative integers. This implies that the dynamics of process X_t, $t \geq 0$ can be expressed as:

$$X_{t+1} = X_t + d - Q\mathbb{1}_{\{X_t \geq R\}}, \tag{1}$$

where $\mathbb{1}_{\{A\}}$ is the indicator function, which is equal to 1 if A holds and is nil otherwise, and d is a value drawn from f_D that represents the surgical demand in period t. The state space of process X_t, $t \geq 0$ is given by $S = \{0, 1, 2, \ldots\}$, where $X_t = i$ indicates that there are $R - Q + i$ patients waiting for elective surgery.

Letting

$$Y_t = \left\lfloor \frac{X_t}{Q} \right\rfloor, \qquad Z_t = X_t \,\%\, Q, \tag{2}$$

where % represents the modulus operator, we can write:

$$X_t = QY_t + Z_t,$$

where Y_t, $t \geq 0$ is the integer number of batches required to bring the queue instantaneously below R and $Z_t \in \{0, 1, \ldots, Q - 1\}$ is the number of patients that would remain in excess of the minimum allowed queue of $(R - Q)$ patients after removing Y_t batches of Q patients instantaneously.

Note that, from the definitions above, the state space of process $Y_t, t \geq 0$ is $S_Y = \mathbb{Z}_+$, while $S_Z = \{0, 1, \ldots, Q - 1\}$ is the state space of process Z_t, $t \geq 0$. In the next result, we will derive the long-term distribution of process Z_t, $t \geq 0$.

Proposition 1. *Let $\pi_Z : S_z \to [0, 1]$ be the steady state distribution of process X_t, $t \geq 0$. Then, it holds that*

$$\pi_Z(z) = \frac{1}{Q}, \forall z \in S_Z. \tag{3}$$

Proof. The proof closely follows that of Axsäter (2015, Proposition 5.1). Let P_Z be the transition matrix describing the dynamics of process Z_t, $t \geq 0$. Assume that $Z_t = z$, $z \in S_Z$ at the start of period $t \geq 0$, and let and $D = k$ be the number of new patient arrivals in period t. Hence, it follows that

$$Z_{t+1} = (z + k) \,\%\, Q = (z + k + QY_t) \,\%\, Q = (X_t + k) \,\%\, Q,$$

where the last equality come from the definitions in (2). Therefore, given $D = k$, it is evident that $p^Z_{zz'}(k) := P(Z_{t+1} = z'|Z_t = z, D = k)$ equals one for exactly one value of z and zero otherwise. Consequently, we can write:

$$\sum_{z=0}^{Q-1} p^Z_{zz'} = \sum_{z=0}^{Q-1} E_k\left\{p^Z_{zz'}(k)\right\} = E_k\left\{\sum_{z=0}^{Q-1} p^Z_{zz'}(k)\right\} = E_k\{1\} = 1.$$

The expression above implies that the Markov chain describing the dynamics of process Z_t, $t \geq 0$ is doubly stochastic, and therefore has a uniform steady state distribution. Indeed, substituting (3) in the steady state equations:

$$\pi^Z(z) = \sum_{z'=0}^{Q-1} \pi^Z(z') p^Z_{z'z},$$

one can easily see that the steady state distribution satisfies Eq. (3). $\square$

The result in Proposition 1 is very important, as it shows that the steady state distribution of process Z_t, $t \geq 0$, is uniform regardless of the distribution of patient arrivals. Next, we will use this fact to derive the steady state distribution of process Y_t, $t \geq 0$.

3.2. The dynamics of process Y_t, $t \geq 0$

Assume that process Z_t, $t \geq 0$ has reached equilibrium, which is also equivalent to supposing that $P(Z_0 = z) = \pi_Z(z) = \dfrac{1}{Q}$ (e.g., Brémaud, 2020). We can describe the dynamics of process Y_t, $t \geq 0$ as:

$$Y_{t+1} = \begin{cases} Y_t - 1 + A, & \text{if } Y_t \geq 0, \\ A; & \text{if } Y_t = 0, \end{cases} \tag{4}$$

where A is the round number of batch arrivals at time t. To fully characterize the random variable A, it is necessary to define

$$A(z) = \left\lfloor \frac{D + z}{Q} \right\rfloor, \tag{5}$$

which represents the round number of batch arrivals at time t, given that $Z_t = z$. Then, in view of Proposition 1, it follows that:

$$P(A = k) = \frac{1}{Q} \sum_{z=0}^{Q-1} P(A(z) = k). \tag{6}$$

Therefore, from (4) we have:

$$p^Y_{ij} = P(Y_{t+1} = j | Y_t = i) = \begin{cases} P(A = j), & \text{if } Y_t = 0, \\ P(A = j - i + 1), & \text{if } Y_t \geq 1. \end{cases} \tag{7}$$

Note that Eq. (4)-(7) correspond to the dynamics in Shortle et al. (2018, Section 6.1.2). Similarly to Shortle et al. (2018), the transition matrix of process Y_t, $t \geq 0$ can be written as:

$$P^Y = [p_{ij}] = \begin{bmatrix} k_0 & k_1 & k_2 & \dots & k_M & 0 & 0 & \dots \\ k_0 & k_1 & k_2 & \dots & k_M & 0 & 0 & \dots \\ 0 & k_0 & k_1 & k_2 & \dots & k_M & 0 & \dots \\ 0 & 0 & k_0 & k_1 & k_2 & \dots & k_M & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \end{bmatrix}, \tag{8}$$

where M is the maximum number of new batch arrivals, i.e., the maximum value that random variable A can assume.

The steady state probabilities of process Y_t, $t \geq 0$ can then be recursively evaluated by solving:

$$\pi_Y^T = \pi_Y^T P, \tag{9}$$

with the contour condition that $\pi_Y(0) = 1 - \rho$ (Shortle et al., 2018, Eq. (6.12)), where $\rho = E(A)$. Observe that, to guarantee stability, we must have $E(A) < 1$. Note also that the steady state probabilities up to a given state j can be recursively calculated using the first j equations in (9).

3.3. Upper and lower bounds on the batch size Q under waiting time requirements

Let ω be the waiting time of a patient joining the queue, and assume that the healthcare system requires that:

$$P(\omega > \tau) < \alpha, \tag{10}$$

for some target waiting time $\tau > 0$ and probability threshold $\alpha \in (0, 1)$. Now, if a patient arrives and encounters $Y_t > 0$ batches of patients in the queue, their turn will only arrive once these Y_t batches are processed, which will take Y_t time periods. This means that the patient will have to wait $\tau = Y_t$ periods for their turn. Therefore, Eq. (10) can be equivalently stated as:

$$\sum_{i=0}^{\tau} \pi_Y(i) > 1 - \alpha, \tag{11}$$

where $\pi_Y : S_Y \to (0, 1)$ satisfies Eq. (9). Therefore, to limit the waiting time according to the prescribed requirements, we can establish a lower bound ($\underline{Q}$) on the batch size $Q > 0$ as follows:

$$\underline{Q} = \min\{Q \in \mathbb{N} : \sum_{i=0}^{\tau} \pi_Y(i) > 1 - \alpha\}. \tag{12}$$

When $Y_t = 0$, on the other hand, the next batch will only be processed when the system leaves this state, therefore by applying the strong Markov property (Brémaud, 2020), we have:

$$\omega = \min\{\delta > 0 : Y_\delta > 0 | Y_0 = 0\}.$$

Considering the transition matrix in (7), it follows that ω will be a geometrically distributed variable and:

$$P(\omega > \tau | Y_0 = 0) = k_0^\tau. \tag{13}$$

One can intuitively see that the value of k_0 will increase as Q increases, since in this case we will need more patient arrivals until we are able to process a full batch of patients. Hence, from Eq. (13), we can set up an upper bound $\overline{Q}$ for the batch size such that:

$$\overline{Q} = \max\{Q \in \mathbb{N} : k_0^\tau < \alpha\}. \tag{14}$$

Therefore, considering the set of specialties $\mathcal{S}$, we can calculate batch sizes $\overline{Q}_s$ and $\underline{Q}_s$ for every specialty $s \in \mathcal{S}$, and then ensure that the waiting time requirements are satisfied both when patients arrive to an empty queue and when they encounter batches of patients in the queue. In the example below, we show how to derive general bounds $\underline{Q}$ and $\overline{Q}$ for a given demand distribution and waiting time requirements.

Example 1 (Deriving lower and upper bounds for Q)**.** *Consider an elective surgery queue with monthly demand D that satisfies the distribution in Table 1.*

Table 1: Monthly demand distribution for surgeries

Demand (d)	0	1	2	3	4	5	6	7	8	9	10
$P(D = d)$	0.0308	0.0393	0.1179	0.2096	0.2445	0.1956	0.1086	0.0414	0.0103	0.0015	0.0005

The expected monthly demand is therefore $E(D) = 3.9022$. Setting $Q = 4$, which ensures long-term stability, we have $S_Z = \{0, 1, 2, 3\}$. This allows us to evaluate the individual distributions of $A(z)$, $z \in S_Z$ as defined in Eq. (5)*, which we depict in the Table 2, along with the distribution of variable A from Eq.* (6)*.*

The distributions in Table 2 follow directly from Eq. (5) *and Eq.* (6)*. For example, when $z = 0$, we know that the queue comprises an integer number of batches. Therefore, $P(A(0) = 0) = p_D(0) + p_d(1) + p_D(2) + p_D(3)$ (Table 1) as up to 3 new arrivals will not suffice to generate a new batch of $Q = 4$ patients. Analogously, $A(0) = 1$ if we observe from 5 to 7 arrivals and $A(0) = 2$ when 8 to 10 patients arrive in the period, as these will suffice for 2 new batches.*

Table 2: Monthly demand distribution for surgeries

$P(A(z) = d)$	Batch demand (d)			
	0	1	2	3
$z = 0$	0.3976	0.5901	0.0123	0
$z = 1$	0.1880	0.7583	0.0537	0
$z = 2$	0.0701	0.7676	0.1618	0.0005
$z = 3$	0.0308	0.6113	0.3559	0.002
$P(A = d)$	**0.171625**	**0.681825**	**0.145925**	**0.000625**

Conversely, when $z = 3$, we have three patients in excess of a discrete number of batches, which means that $P(A(3) = 0) = p_D(0)$ as new batches will be formed whenever one or more patients arrive. One new batch will form if $D \in [1, 4]$, two new batches if $D \in [5, 8]$ and three new batches when $D \in [8, 10]$. Specifically, when 10 patients arrive, they will form a set of 13 patients when joined to the $z = 3$ remaining patients from the start of the period. These 3 patients will then suffice to form three batches of $Q = 4$ patients, with one remaining batchless for the following period.

Note that $E(A) = 0.9755$, and that this value is equal to $\rho = \frac{E(D)}{Q} = \frac{3.9022}{4} = 0.9755$. This is expected, as the dynamics of process Y_t describes the original system, only looking at it in terms of batches of patients. Note that $E(A)$ is simply the expected number of batches of Q patients arriving and since we process $\mu = 1$ batch per time period, we have $\rho = E(A)$ when looking at the evolution of the batch process Y_t, $t \geq 0$.

Now, let us suppose that $\tau = 6$ and $\alpha = 0.05$ in (10), which means that the system requires 95% of the patients to be operated in up to six months. To verify whether that is satisfied, we need firstly to solve Eq. (9) for the steady state distribution of Y_t, $t \geq 0$ and then use it to verify (11). From Table 2, we obtain $k_0 = 0.171625$, $k_1 = 0.681825$, $k_2 = 0.145925$ and $k_3 = 0.000625$, which can be applied in Eq. (9) to find $\pi_Y(0) = 0.02445$, $\pi_Y(1) = 0.1182$, $\pi_Y(2) = 0.1219$, $\pi_Y(3) = 0.1046$, $\pi_Y(4) = 0.0898$, $\pi_Y(5) = 0.077$, $\pi_Y(6) = 0.0661$. This yields:

$$P(\omega \leq 6 | Y_t \geq 1) \approx 0.602 < 0.95 = 1 - \alpha.$$

This implies that $\underline{Q} > 4$ in Eq. (12) and we should, therefore, recalculate the probability above for increasing values of Q until we find the first value that satisfies the waiting time criterion - which will give us the lower bound $\underline{Q}$.

To verify the waiting time requirements when $Y_t = 0$, one can apply (13), which implies that $P(\omega \leq 6 | Y_t = 0) = 0.171625^6 \approx 2.5 \cdot 10^{-5} < 0.05$. Therefore, to find the bound $\overline{Q}$ in Eq. (14).

When verifying Eq. (12) with $Q = 10$, it implies $P(\omega \leq 6 | Y_t = 0) = 0.609780^6 \approx 0.05114 > 0.05$. Then, if one set $Q = 9$, it implies $P(\omega \leq 6 | Y_t = 0) = 0.566478^6 \approx 0.03304 < 0.05$. Thus, we can set $\overline{Q} = 9$.

For the $\underline{Q}$, one can verify $Q = 5$, which implies $k_0 = 0.265720$, $k_1 = 0.688120$, $k_2 = 0.046160$ and $k_3 = 0$, which can be applied in Eq. (9) to find $\pi_Y(0) = 0.21956$, $\pi_Y(1) = 0.606723$, $\pi_Y(2) = 0.143539$, $\pi_Y(3) = 0.024935$, $\pi_Y(4) = 0.004332$, $\pi_Y(5) = 0.000752$, $\pi_Y(6) = 0.000131$. This yields:

$$P(\omega \leq 6 | Y_t \geq 1) \approx 0.999973 > 0.95 = 1 - \alpha.$$

Thus, we can set $\underline{Q} = 5$.

4. Estimating the number of patients per time block

To ensure consistency with the waiting-time requirements established by the (R, Q) policy in Section 3, it is necessary to estimate how many elective patients can be accommodated within a single 6-hour operating theatre block, in accordance with the scheduling regulations at HC-Unicamp. This estimation determines whether the batch sizes $\overline{Q}_s$ and $\underline{Q}_s$, for each specialty $s \in \mathcal{S}$, defined at the strategic level can be effectively served within the planning horizon.

Accurate estimation requires explicit consideration of uncertainty in surgery durations, which varies across sub-specialties and directly affects both operating theatre utilisation and the risk of overtime. In particular, a trade-off arises between idle time, resulting from underutilised theatre capacity, and overtime, which may trigger cancellations of scheduled procedures.

Table 3: Table of Symbols for Overtime and cancellation management process.notaçao entre seçoes está caótica, vou sincronizar tudo no symb.tex.

Index	Description
i	Surgery index
k	Number of completed surgeries
u	Number of cancellations
Parameter	**Description**
T	Length of the surgery block (hours)
t^*	Safety time threshold required to start a new surgery, resulting from the Newsvendor model
$\varepsilon_{\text{stop}}$	Probability threshold used as stopping criterion in the algorithm
n	Number of surgeries planned for a block of length T
p	Discretised probability mass function of a single surgery duration
Random Variable	**Description**
d_i	Duration of the i-th surgery
s_k	Cumulative duration of the first k surgeries, $s_k = \sum_{i=1}^{k} d_i$
X	Number of surgeries performed within the block
Function	**Description**
$F_{s_k}(\cdot)$	Cumulative distribution function of s_k
Convolve$(\cdot,\cdot)$	Convolution operator used to compute cumulative duration distributions
RunNewsvendor$(\cdot)$	Function to compute $P(s_k < T - t^*)$
$G(\cdot)$	Gain function associated with scheduling additional surgeries
$C(\cdot)$	Cost function associated with cancellations
$\Delta(\cdot)$	Net value function combining gain and cancellation cost
$R(\cdot)$	Marginal value function for scheduling additional surgeries
$c(\cdot)$	Function to compute cancellation probabilities
Derived Quantity	**Description**
$P(X > k)$	Probability that more than k surgeries can be completed (survival function)
$P(X = k)$	Probability that exactly k surgeries can be completed
$K_{\max}$	Maximum number of surgeries considered, such that $F_{s_k}(T - t^*) > \varepsilon_{\text{stop}}$
$q[k]$	Discrete representation of $P(X > k)$ (survival function of X)
$c^{(k)}$	Probability distribution of cumulative duration s_k
Output Metric	**Description**
$P(\text{cancellations} = \ell)$	Probability that ℓ surgeries are cancelled

For this purpose, in this section we present the second step of the proposed integrated framework. A newsvendor-based model is first employed to determine an optimal time allocation for the final surgery in a block under duration uncertainty, balancing idle-time and overtime costs. Building on this result, an inventory-based formulation is then used to estimate the number of patients that can be reliably scheduled within a time block, providing the tactical input required for subsequent operating theatre allocation decisions. The symbols used in this section are defined in Table 3.

4.1. Managing surgery cancellations via a newsvendor-based model

Let us now consider the problem of determining the optimal time budget for the last surgery in a session, based on the random duration of this surgery. Let T_D be a positive random variable taking values from $\Omega \in \mathbb{R}_+$, which represents the duration of a given surgery, with cumulative distribution $F_{T_D} : \Omega \to [0, 1]$. Assume that, once the theatre is ready for the surgery, it only proceeds as planned if the remaining time in the session exceeds a prescribed budget of $t_b \in \Omega$ time units. Otherwise, the surgery is called off.

The time budget $t_b \in \Omega$ can be interpreted as a perishable inventory of session time. If the inventory is excessive, we will not fully utilise the time allocated to the surgical team to perform the surgery. Conversely, if we proceed with the surgery, but the time budget is insufficient, the surgical team will have to work overtime to complete the surgery. Let $c > 0$ be the cost of each unit of time budget in excess of the actual surgical time T_D and $p > c$ be a penalty for each unit of overtime that the surgical team incurs. The expected cost of the time deviation between T_D and t_b is defined as:

$$u(t_b) = cE(\beta_1) + pE(\beta_2), \tag{15}$$

where $\beta_1 = \min\{t_b - T_D,\ 0\}$ is the underused time budget, and $\beta_2 = \max\{T_D - t_b,\ 0\}$ is the total overtime. The objective is to find an optimal time budget allocation t^* that minimizes the time deviation cost in Eq. (15), which is given by:

$$t^* = \arg\min_{t_b \in \Omega} u(t_b). \tag{16}$$

Lemma 1. *Let* $\beta_1 = \min\{t_b - T_D,\ 0\}$ *and* $\beta_2 = \max\{T_D - t_b,\ 0\}$*. Then, it follows that:*

$$E(\beta_1) = \int_0^{t_b} F_{T_D}(t)dt, \quad E(\beta_2) = \int_{t_b}^{\infty} (1 - F_{T_D}(t))dt \tag{17}$$

Proof. We will start by proving the result for β_1. Since $\beta_1 \geq 0$, it follows that:

$$E(\beta_1) = \int_{t=0}^{\infty} P(\beta_1 > t)dt = \int_{t=0}^{t_b} P(t_b - T_D > t)dt = \int_{t=0}^{t_b} P(T_D < t_b - t)dt = \int_{t=0}^{t_b} F_{T_D}(t_b - t)dt = \int_{t=0}^{t_b} F_{T_D}(t)dt.$$

Since β_2 is also non-negative, we can write:

$$E(\beta_2) = \int_0^{\infty} P(T_D - t_b > t)dt = \int_0^{\infty} P(T_D > t + t_b)dt = \int_{t_b}^{\infty} P(T_D > t)dt = \int_{t_b}^{\infty} (1 - F_{T_D}(t))dt$$

□

Theorem 1. *Let* $u(t_b)$ *be defined in Eq.* (15)*, and* $t*$ *be a solution to Eq.* (16)*. Then, if follows that:*

$$t^* = F_{t_D}^{-1}\left(\frac{\frac{p}{c}}{1 + \frac{p}{c}}\right), \tag{18}$$

where $F_{T_D}^{-1}(\cdot)$ *is the inverse of the cumulative distribution function* $F_{T_D}(\cdot)$*.*

Proof. Substituting (17) in Eq. (15) yields:

$$u(t_b) = c\int_0^{t_b} F_{T_D}(t)dt + p\int_{t_b}^{\infty} (1 - F_{T_D}(t))dt.$$

When $t_b = t^*$, the first order optimality condition for $u(t_b)$ implies:

$$\frac{du(t_b)}{dt_b} = cF_{T_D}(t^*) - p(1 - F_{T_D}(t^*)) = 0.$$

Hence, we must have:

$$cF_{T_D}(t^*) - p(1 - F_{T_D}(t^*)) \implies \frac{F_{T_D}(t^*)}{(1 - F_{T_D}(t^*))} = \frac{p}{c} \implies F_{T_D}(t^*)\left(1 + \frac{p}{c}\right) = \frac{p}{c}$$

We conclude the proof by noting that (18) follows from the expression above. □

Theorem 1 implies that, whenever the remaining time in a session exceeds the quantile t^* in Eq. (18), the surgery is greenlighted to continue. Otherwise, the surgery is cancelled and the session comes to a close. Note that the quantile is solely a function of the ratio between the overtime cost p and the cost of unused session time c, and is applicable to any surgery time distribution. For example, if $p = 2c$, $t^* = F_{T_D}^{-1}\left(\frac{2}{3}\right)$. This means that at least two thirds of the green-lighted surgeries will not require overtime.

4.2. Applying the newsvendor model to the case study

Building on the theoretical results of the previous subsection, we evaluate, for a block of length $T \in \mathbb{Z}_+$, the probability that a further surgery can be accommodated after $k \in \mathbb{N}$ surgeries have been completed. This is equivalent to assessing whether the residual time in the block exceeds the safety threshold t^*. As described below, this probability is obtained through successive convolutions of the surgery-duration distributions associated with the relevant subspecialty.

Let d_i denote a non-negative the random variable representing the duration of the i-th surgery, with probability distribution $f_{d_i} : \Omega \to [0, 1)$, $\Omega \subset \mathbb{R}_+$ and cumulative distribution function $F_{d_i} : \Omega \to [0, 1)$. Let $s_k = \sum_{i=1}^{k} d_i$ denote the cumulative duration of the first k surgeries. The probability that an additional surgery can be performed after k procedures is given by

$$\begin{aligned} P(T - s_k > t^*) &= P(s_k < T - t^*) \\ &= F_{s_k}(T - t^*), \end{aligned}$$

where $F_{s_k} : \Omega \to [0, 1)$, $\Omega \subset \mathbb{R}_+$ is the cumulative distribution function of s_k.

We also define the random variable X as the non-negative number of surgeries that can be executed within the block, with probability distribution $f_X : \Omega \to [0, 1)$, $\Omega \subset \mathbb{N}_+$ and cumulative distribution function $F_X : \Omega \to [0, 1)$. Therefore, the probability that more than k surgeries can be performed is given by

$$P(X > k) = P(s_k < T - t^*).$$

Preciso rever a explicacao do criterio de parada, porque do jeito em que está, eu explico o K_{max} mas nem uso ele, só a ideia... Rever a explicação e o pseudocódigo tbm. To ensure computational tractability, we define an upper bound $K_{\max}$ as the largest integer such that

$$K_{\max} = \max \left\{ k \geq 0 : F_{s_k}(T - t^*) > \varepsilon_{\text{stop}} \right\},$$

where $\varepsilon_{\text{stop}} \in (0, 1)$ is a pre-specified probability threshold. This definition avoids ambiguity by directly linking $K_{\max}$ to the stopping criterion used in the algorithm.

Given the model parameters $T, t^*, \varepsilon_{\text{stop}}$, and the discretised probability mass function of surgery durations p_s for all $s \in \mathcal{S}$, we compute, for each subspecialty, the probabilities $P(X > k)$ for $k = 0, \ldots, K_{\max}$ iteratively using convolution operations, as described in Algorithm 1.

Algorithm 1 Computation of $P(X > k)$ via convolutions and the newsvendor model

Input: T, t^*, $\varepsilon_{\text{stop}}$, specialties $s \in \mathcal{S}$ and probability vectors p_s for all $s \in \mathcal{S}$
Output: vector q such that $q[k] = P(X > k) = P(s_k < T - t^*)$
for each $s \in \mathcal{S}$ **do**
  Initialise $q[0] \leftarrow 1$
  Initialise $c^{(0)} \leftarrow p_s$
  $k \leftarrow 0$
  while $q[k] > \varepsilon_{\text{stop}}$ **do**
    if $k > 0$ **then**
      $c^{(k)} \leftarrow \text{Convolve}(c^{(k-1)}, p_s)$
    end if
    $q[k+1] \leftarrow \text{RunNewsvendor}(c^{(k)}, T, t^*)$
    $k \leftarrow k + 1$
  end while
end for

In Algorithm 1, the function `Convolve` computes the distribution of the cumulative duration s_k by convolving the distribution of s_{k-1} with the single-surgery duration distribution p_s. The function `RunNewsvendor` evaluates $P(s_k < T - t^*)$, i.e., the probability that an additional surgery can be accommodated.

The resulting vector $q[k]$ provides the survival function $P(X > k)$. The probability mass function of X can then be obtained by

$$P(X = k) = P(X > k - 1) - P(X > k), \quad k \geq 1. \tag{19}$$

These probabilities allow us to characterise the distribution of the number of surgeries that can be completed within a block.

Finally, from Equation 19, we can derive the probability of cancellations for a given number of surgeries planned in advance. Let n denote the number of scheduled surgeries within a block. If X surgeries can be performed, then $n - X$ surgeries will be cancelled. Therefore, the probability of cancelling ℓ surgeries is

$$P(\text{cancellations} = \ell) = P(X = n - \ell), \quad \ell = 0, 1, \ldots, n - 1. \tag{20}$$

Overall, the proposed newsvendor-based model enables the estimation of both the number of surgeries that can be performed within a block and the associated cancellation probabilities. These quantities are essential for supporting informed scheduling decisions. In particular, they provide the basis for determining the optimal number of surgeries to schedule within a block, as discussed in the next subsection.

4.3. Inventory-based model for defining the number of surgeries in a time block

TODO: ADD CITATIONS ABOUT INVENTORY MODEL USED FOR MODELING HERE

Using the probability distributions derived in the previous subsection, we now determine the number of surgeries n to be planned within a single operating theatre block. This decision is formulated through an inventory-based perspective that balances the benefits of scheduling additional surgeries against the risks associated with cancellations.

In contrast to classical inventory formulations, where the gain is typically proportional to the expected number of served units, we adopt an alternative interpretation of the benefit associated with performing surgeries. Specifically, we define the gain of planning n surgeries as

$$G(n) = R \cdot P(X > n - 1), \tag{21}$$

where $R > 0$ is a constant representing the marginal value of scheduling an additional surgery.

In this context, $G(n)$ should be interpreted as a marginal gain associated with doing more surgeries in a block, rather than as a direct monetary return. This interpretation is particularly relevant in public healthcare settings, where the objective is not purely financial. Instead, the value of scheduling additional surgeries may reflect multiple factors, such as improved access to care, reduction of waiting lists, and better utilisation of operating theatre capacity. However, the marginal benefit of adding further surgeries typically decreases, with various factors contributing to this effect, such as team fatigue, increased risk of complications, and diminishing returns in terms of patient outcomes.

The cost associated with cancellations is modelled through a function $c(\ell)$, where ℓ is the number cancelled surgeries. We assume that $c(\cdot)$ is a non-decreasing function with $c(0) = 0$, reflecting the fact that cancellations generate administrative, logistical, and patient-related costs.

The expected cancellation cost when planning n surgeries is given by

$$C(n) = \mathbb{E}[c((n - X))] = \sum_{x=0}^{n} P(X = x) \cdot c(n - x). \tag{22}$$

The analogous net value of planning n surgeries is therefore defined as

$$\Delta(n) = G(n) - C(n). \tag{23}$$

The optimal number of surgeries to be planned within a block is obtained by solving

$$n^* = \max\{n : \Delta(n) > 0\}.$$

In this manner, we choose the probability distribution of X that maximises the net value $\Delta(n)$, which captures the trade-off between the benefits of scheduling additional surgeries and the costs associated with potential cancellations.

This formulation explicitly captures the trade-off between increasing the likelihood of performing additional surgeries and the risk of cancellations. The optimal value n^* depends on the distribution of X, derived from the surgery-duration distributions, as well as on the parameters R and $C(\cdot)$.

For each subspecialty, the optimal number of surgeries to be planned within a block can be determined by applying this model with the corresponding duration distributions. These estimates are then used as input for subsequent operating theatre allocation decisions, as described in the next subsection.

5. Operating Theatre Scheduling Problem

Following the proposed integrated framework, the final step consists of solving the Operating Theatre Scheduling Problem (OTSP), which addresses the tactical allocation of subspecialties to operating theatre blocks. This stage is essential for translating the strategic decisions derived from the preceding layers into actionable schedules that can be implemented in hospital operations. To this end, the model takes as input the demand for each subspecialty, driven by the (R, Q) policy, as well as the effective capacity of each block, determined in the cancellation and overtime management stage.

We formulate the OTSP as a Mixed-Integer Linear Programming (MILP) model, which enables the representation of complex constraints and multiple, potentially competing objectives, while ensuring optimality with respect to the specified criteria. The model incorporates several objectives aligned with hospital priorities, including maximising operating theatre utilisation, ensuring the allocation of priority subspecialties, and promoting efficient sequencing of block allocations from a logistical perspective.

Nevertheless, the proposed framework is sufficiently flexible to accommodate alternative solution approaches, such as heuristics or metaheuristics. This flexibility arises because the outputs of the previous layers, namely demand and effective capacity, can serve as inputs to any scheduling methodology, not exclusively the MILP formulation proposed here. This highlights the value of integrating the different decision layers, as it provides a structured mechanism for incorporating both strategic and operational insights into the scheduling process, irrespective of the specific solution approach adopted for the OTSP.

In the following subsections, we first describe the preprocessing of strategic inputs derived from the (R, Q) policy and the effective capacity information in order to determine subspecialty demand in terms of block units, which constitute the relevant unit for the scheduling problem. We then present the MILP formulation of the OTSP, detailing the decision variables, objective functions, and constraints that reflect the hospital's operational requirements and priorities.

5.1. Preprocessing strategic input

The OTSP requires demand to be expressed in block units, as these correspond to the operational granularity of theatre scheduling. To this end, we preprocess the strategic outputs of the (R, Q) policy together with the effective capacity information obtained from the cancellation and overtime management stage.

The overtime and cancellation stage provides the probability mass function p_s of the number of surgeries that can be completed within a single block for subspecialty s. To determine the block requirements associated with targets $\underline{Q}_s$ and $\overline{Q}_s$ at confidence level ρ, we iteratively convolve this distribution.

Let $F_s^{(r)}$ denote the cumulative distribution function associated with the PMF obtained after convolving p_s with itself r times. Then, for a generic target $Q \in \{\underline{Q}_s, \overline{Q}_s\}$, the minimum number of blocks required is the smallest integer r such that

$$1 - F_s^{(r)}[Q - 1] \geq \rho.$$

In other words, we seek the smallest number of blocks r such that the probability of completing at least Q surgeries is at least ρ. Algorithm 2 summarises this procedure.

Algorithm 2 Block demand estimation via successive convolutions

Input: set of subspecialties $\mathcal{S}$, targets $(\underline{Q}_s, \overline{Q}_s)$, confidence level ρ, PMF p_s of surgeries per block for each $s \in \mathcal{S}$
Output: block requirements $(\underline{B}_s, \overline{B}_s)$ for each $s \in \mathcal{S}$
for each subspecialty $s \in \mathcal{S}$ **do**
  if $\underline{Q}_s = 0$ and $\overline{Q}_s = 0$ **then**
    $\underline{B}_s \leftarrow 0, \overline{B}_s \leftarrow 0$
  else
    for each $Q \in \{\underline{Q}_s, \overline{Q}_s\}$ **do**
      $counter \leftarrow 1$
      $p_s^{(counter)} \leftarrow p_s$
      $F \leftarrow \text{CalculateCDF}\left(p_s^{(counter)}\right)$
      while $1 - F[Q - 1] < \rho$ **do**
        $p_s^{(counter+1)} \leftarrow \text{Convolve}\left(p_s^{(counter)}, p_s\right)$
        $counter \leftarrow counter + 1$
        $F \leftarrow \text{CalculateCDF}\left(p_s^{(counter)}\right)$
      end while
      if $Q = \underline{Q}_s$ **then**
        $\underline{B}_s \leftarrow counter$
      else
        $\overline{B}_s \leftarrow counter$
      end if
    end for
  end if
end for

This step is essential for the OTSP phase of the framework, since the problem is formulated at the block level and the monthly operating theatre schedule is constructed by allocating blocks rather than individual surgeries. Moreover, this preprocessing step enables the uncertainty captured in the strategic layer to be incorporated into the framework without introducing stochastic elements directly into the scheduling model. This simplifies both the model formulation and the solution process (i.e., the OTSP is solved as a deterministic MILP), while ensuring that the resulting schedules remain aligned with demand targets that reflect both variability in surgical durations and the effective capacity of blocks.

5.2. *MILP Formulation*

We model the OTSP as a MILP, which allows us to capture the complex interactions between subspecialty demand, block capacity, and scheduling constraints in a structured manner. To formulate the problem, we use the the notation in Table 4.

The Objective Functions (OFs) are organised according to the priority levels established by the hospital's guidelines – although they can be reordered as needed to reflect different institutional priorities in other settings. To reflect these priorities, we adopt a hierarchical optimisation strategy, as proposed by Aringhieri et al. (2022). The OFs are optimised sequentially, while respecting the optimal values obtained in the previous stages.

Formally, the first OF is defined as

$$\text{Min } F_0 = \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} z_{sb} + C \cdot a_b, \tag{24}$$

which minimises the total slack in team and anesthesiologist availability, with a high penalty on the latter as they are more scarce and expensive. The second function is given by

$$\text{Min } F_1 = \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{tbs} \text{ only for variables where } \lambda_{tbs} = 0 \tag{25}$$

which minimises the number of allocations that violate the usual availability of rooms for each subspecialty, thus promoting the use of preferred rooms and reducing the need for last-minute adjustments. The third objective function is defined as

$$\text{Max } F_2 = \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}_{prio}} x_{tbs}, \tag{26}$$

Table 4: Table of Symbols notaçao entre seçoes está caótica, vou sincronizar tudo no symb.tex.

Set	Description
$\mathcal{T}$	Operating theatres (both elective and outpatient)
$\mathcal{D}$	Days of the planning horizon, such that $\mathcal{D} = \{1, 2, ..., \lvert\mathcal{D}\rvert\}$
$\mathcal{M}$	Periods (morning or afternoon), such that $\mathcal{M} = \{\text{morning}, \text{afternoon}\}$
$\mathcal{B}$	Blocks, such that $\mathcal{B} = \mathcal{D} \times \mathcal{M}$
$\mathcal{S}$	Surgical subspecialties
$\mathcal{S}_{seq}$	Surgical subspecialties that necessarily require two sequential allocated blocks, such that $\mathcal{S}_{seq} \subset \mathcal{S}$ but have high priority, such that $\mathcal{S}_{circ} \subset \mathcal{S}$
$\mathcal{S}_{prio}$	Priority surgical subspecialties, such that $\mathcal{S}_{prio} \subset \mathcal{S}$
$\mathcal{S}_{trans}$	Organ transplant surgical subspecialties, such that $\mathcal{S}_{trans} \subset \mathcal{S}$
Index	**Description**
t	Operating theatre, with $t \in \mathcal{T}$
d	Planning horizon day, with $d \in \mathcal{D}$
m	Period (morning or afternoon), with $m \in \mathcal{M}$
b	Surgery block, with $b \in \mathcal{B}$
s	Surgical subspecialty, with $s \in \mathcal{S}$
Parameter	**Description**
$\overline{B}_s$	Maximum number of blocks allocated to each subspecialty $s \in \mathcal{S}$
$\underline{B}_s$	Minimum number of blocks allocated to each subspecialty $s \in \mathcal{S}$
η_{bs}	Number of teams of subspecialty $s \in \mathcal{S}$ available in block $b \in \mathcal{B}$
A_b	Number of anesthesiologists available in block $b \in \mathcal{B}$
γ_s	1 if subspecialty $s \in \mathcal{S}$ requires an anesthesiologist; 0 otherwise
β_s	Minimum number of blocks for subspecialty $s \in \mathcal{S}$
C	Constant to penalize the anesthesiologist slack variable a_b in the objective function ($C = \lvert\mathcal{B}\rvert \cdot \lvert\mathcal{S}\rvert \cdot \lvert\mathcal{T}\rvert$)
λ_{tbs}	1 if room $t \in \mathcal{T}$, in block $b \in \mathcal{B}$, can be used by subspecialty $s \in \mathcal{S}$; 0 otherwise
Variable	**Description**
x_{tbs}	1 if room $t \in \mathcal{T}$, in block $b \in \mathcal{B}$, is allocated to subspecialty $s \in \mathcal{S}$; 0 otherwise. This variable is only created if the allocation is feasible (i.e., specialty s is compatible with the theatre t).
y_{tds}	1 when room $t \in \mathcal{T}$, on day $d \in \mathcal{D}$, is allocated only to subspecialty $s \in \mathcal{S}$; 0 otherwise
w_{bs}	Number of transplants to be performed in block $b \in \mathcal{B}$ by subspecialty $s \in \mathcal{S}$
a_b	Slack variable for the number of anesthesiologists available in block $b \in \mathcal{B}$
z_{sb}	Slack variable for the number of teams available in block $b \in \mathcal{B}$ for subspecialty $s \in \mathcal{S}$

which maximises the allocation of priority subspecialties, ensuring that they receive as much capacity as possible within the constraints of the system. The fourth OF is given by

$$\text{Max } F_3 = \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{tbs}, \tag{27}$$

which maximises the total number of allocated blocks, thus promoting overall capacity utilisation and service provision. Finally, the fifth OF is defined as

$$\text{Max } F_4 = \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{tbs} \tag{28}$$

which maximises the number of sequentially allocated blocks. This is particularly important for improving the logistics of the surgical process, reducing the need for equipment and team transfers between theatres.

With this, we formulate the theatre planning problem at HC - Unicampwith the model MILP-OT, incorporating these OFs and constraints based on the hospital's current operational framework, but with the flexibility to accommodate other settings as needed.

$$
\begin{aligned}
(MILP-OT) \quad & \text{Optimize } F_0\text{–}F_5 && \\
& \text{subject to} \quad \sum_{t\in\mathcal{T}}\sum_{b\in\mathcal{B}} x_{tbs} \le \overline{B}_s && \forall\, s\in\mathcal{S} && (29)\\
& \sum_{t\in\mathcal{T}}\sum_{b\in\mathcal{B}} x_{tbs} \ge \underline{B}_s && \forall\, s\in\mathcal{S} && (30)\\
& \sum_{s\in\mathcal{S}} x_{tbs} \le 1 && \forall\, t\in\mathcal{T},\ b\in\mathcal{B} && (31)\\
& \sum_{t\in\mathcal{T}}\sum_{s\in\mathcal{S}} \gamma_s \cdot x_{tbs} \le A_b + a_b && \forall\, b\in\mathcal{B} && (32)\\
& \sum_{t\in\mathcal{T}} x_{tbs} \le \eta_{bs} + z_{sb} && \forall\, b\in\mathcal{B},\ s\in\mathcal{S} && (33)\\
& x_{tbs} + x_{t(b+1)s} - 1 \le y_{tds} && \forall\, t\in\mathcal{T},\ b\in\mathcal{B},\ s\in\mathcal{S}, && \\
& && d\in\mathcal{D}:\ \ b \text{ and } b+1 \text{ are blocks} && \\
& && \text{of the same day } d && (34)\\
& x_{tbs} + x_{t(b+1)s} \ge 2\cdot y_{tds} && \forall\, t\in\mathcal{T},\ b\in\mathcal{B},\ s\in\mathcal{S}, d\in\mathcal{D}: && \\
& && d\in\mathcal{D}:\ b \text{ and } b+1 \text{ are blocks} && \\
& && \text{of the same day } d && (35)\\
& x_{tbs} = x_{t(b+1)s} && \forall\, t\in\mathcal{T},\ b\in\mathcal{B}',\ s\in\mathcal{S}_{seq}, && \\
& && d\in\mathcal{D}:\ b \text{ and } b+1 \text{ are blocks} && \\
& && \text{of the same day } d && (36)\\
& \sum_{t\in\mathcal{T}} x_{tbs} = 2\cdot w_{bs} && \forall\, b\in\mathcal{B},\ s\in\mathcal{S}_{trans} && (37)\\
& x_{tbs} \in \{0,1\} && \forall\, t\in\mathcal{T},\ b\in\mathcal{B},\ s\in\mathcal{S} && \\
& y_{tds} \in \{0,1\} && \forall\, t\in\mathcal{T},\ d\in\mathcal{D},\ s\in\mathcal{S} && \\
& w_{bs} \in \mathbb{Z}_+ && \forall\, b\in\mathcal{B},\ s\in\mathcal{S} && \\
& a_b \in \mathbb{Z}_+ && \forall\, b\in\mathcal{B} && \\
& z_{sb} \in \mathbb{Z}_+. && \forall\, b\in\mathcal{B},\ s\in\mathcal{S} &&
\end{aligned}
$$

Constraints (29) and (30) impose minimum and maximum bounds on the number of allocations of each subspecialty. This is part of the integration with the strategic layer, as these bounds are determined based on the demand δ_s for each subspecialty, which in turn is influenced by the (R, Q) policy. Constraint (31) ensures that each theatre and time block receives at most one subspeciality. Constraint (32) is a soft constraint that limits the number of allocations that require anaesthetists according to their availability A_b in each block, while allowing for some flexibility through the slack variable a_b. Constraint (33) similarly limits the number of allocations for each subspecialty based on the availability of medical teams η_{bs}, with the slack variable z_{sb} providing flexibility when necessary. Constraints (34) and (35) together guarantee sequential allocation of blocks on the same day when $y_{tds} = 1$. Subspecialities requiring two consecutive blocks are specifically addressed by constraint (36). Constraint (37) ensures that transplant subspecialties are allocated in pairs, reflecting the need for simultaneous operating theatre use for donor and recipient procedures.tion balancing in conjunction with OF (26).

6. Computational Study

We executed a computational study to evaluate the performance of our proposed approach. Although we have theoretical guarantees to ensure strategic and operational stability, we also want to evaluate the practical performance of our approach in a real-world setting. To do so, we use real data from HC - Unicampto perform a data-driven computational study. In this section, we present the experimental setup and the results of our computational study

6.1. Real-world Data Processing and Instances

6.2. Experimental Setup

The experiments were executed on a computer with an Intel Core 7 150U 5.40GHz and 32GB of RAM. We implemented our approach using Python 3.12 for all stages. SCIP 9.2.3 was used as the optimization solver with default settings (together with Pyomo library) to solve the MILP model for OT scheduling.

We simulated the OT scheduling process for a period of 1 year, using the results from the data processing phase (i.e., the R and Q values, as well as the overtime and cancellation costs) to allocate blocks to each specialty and evaluate the resulting waiting times for patients in each specialty. Figure 3 illustrates the long-term simulation process, where we use the results from the data processing phase to perform OT scheduling and evaluate the resulting waiting times for patients in each specialty.

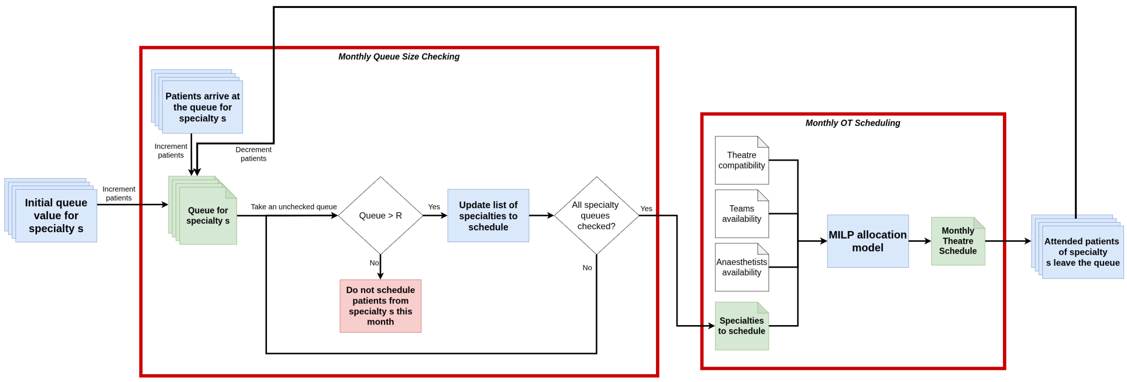

Figure 3: Long-term simulation process.

The experimental workflow is structured in two main stages performed on a monthly basis. First, the queue size of each specialty is evaluated. Starting from the initial queue levels, newly arriving patients are added, resulting in the updated queue for each specialty. Then, each specialty queue is sequentially examined: if its size exceeds the predefined threshold R, the specialty is included in the list of candidates for scheduling; otherwise, no patients from that specialty are scheduled in the current month. This procedure continues until all specialty queues have been checked. In the second stage, the set of selected specialties, together with resource constraints such as theatre compatibility and the availability of surgical teams and anaesthetists, are provided as inputs to a MILP allocation model. The model generates a monthly operating theatre schedule, which determines the number of patients treated in each specialty. Finally, treated patients leave their respective queues, closing the monthly cycle and updating the system state for the next iteration.

To validate the long-term stability, we executed 100 simulations of this period of 1 year, with different random seeds to account for the stochasticity in surgery durations and demand values.

6.3. Data processing Results

6.3.1. (R, Q) *Results*

6.3.2. Overtime and Cancellation Results

6.4. Long-term Stability Impact

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